

Loyalty Signaling in Hong Kong: A Textual Analysis of Political Discourse from 1997 to 2026

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This manuscript was compiled on May 10, 2026.

Abstract

This paper investigates the evolution of “loyalty signaling” by Hong Kong Chief Executives (CEs) toward the central government. While the definition of “loyalty” remains conceptually fluid and vague in Chinese politics, we propose a six-dimensional framework to categorize how political alignment is communicated. And the authors measure the frequency of loyalty signaling by CE through two proxies: (i) *loyalty-signaling keyword density* in CE speeches and press releases, and (ii) the semantic similarity between the speeches of CE and central leaders. The results reveal a distinct cyclical pattern in CE’s loyalty-signaling discourse which intensifies following periods of socio-political friction or “trust crises” between Hong Kong and the Central Government, yet with no evidence suggesting a wholesale importation of Mainland-style rhetorical system into Hong Kong. It is thus suggested that loyalty signaling is employed by CE as a reparative strategy aiming at signaling political alignment to Beijing and regaining its political trust during times of uncertainty and turmoil, so that Hong Kong might gradually recover more room for high autonomy and perhaps get back on track towards democratization as prescribed in the Basic Law.

Keywords: *loyalty signaling, political discourse, central-local relations, text mining*

1. INTRODUCTION

Since the implementation of the Law of the People’s Republic of China on Safeguarding National Security in the Hong Kong Special Administrative Region (‘NSL’) in 2020 and the subsequent top-down overhaul of Hong Kong’s electoral system which produces members of the Legislative Council (‘LegCo’) and the Chief Executive of the HKSAR government (‘CE’) by Beijing, many people have observed a sharp increase of the references to the PRC central authorities, their remarks, their ideology and their policy directions in the SAR government’s public discourse. This phenomenon appears to be increasingly pronounced after the incumbent CE, John Lee, took office in mid-2022. John Lee is viewed by many as proactively engaging the SAR government in a process of “Mainlandization” for the purpose of signaling loyalty to the Central People’s Government (‘CPG’) (Chan, 2023), as opposed to his predecessors, who did not adopt such a “sycophantic” gesture in signaling loyalty to Beijing. Focusing on the discursive aspect of the CE’s loyalty-signaling behaviour, our research aims to explore the extent to which the CE’s public addresses resonate with the language of the CPG and then explore the underlying logic behind their loyalty-signaling behaviours.

This paper will firstly lay out the general theoretical framework on loyalty signaling by lower-level officials to the central authority and the important role of language and discourse amid the loyalty-signaling process by reviewing the existing literature. On this basis, we will conceptualize the phenomenon of “loyalty signaling through language” in the political context of Hong Kong, according to which six types of keywords used to communicate loyalty will be proposed (‘*Loyalty Signaling Keywords*’). We will then quantify the extent to which different CE’s signaled loyalty through discourse by measuring the density of loyalty keywords in CE’s speeches and the CE office’s press releases, with the semantic similarities between the public addresses of CE and central leaders as a further reference that enriches our analysis.

2. GENERAL THEORETICAL FRAMEWORK OF LOYALTY SIGNALING

Qian and Bai (2024) give three fundamental propositions regarding lower-level government officials’ loyalty signaling behaviours towards the central authority in an authoritarian regime: i) officials with a less-trusted background have a greater incentive to signal their loyalty to the authoritarian leader; ii) credible loyalty signaling becomes particularly significant for an official if his or her career prospect and political survival is largely determined by the ruler’s favor, which is often the case in an authoritarian regime; iii) one of the most effective ways for a less-trusted official to express his or her loyalty is to engage in “unpopular and costly” tasks on the ruler’s behalf and for the ruler’s benefit. Whilst Qian and Bai (2024) focus more on how local bureaucrats used political persecution to express loyalty amid the Anti-Rightist Campaign in the early years of the PRC, Baturo and Khokhlov (2025) highlight the local officials’ politicized framing of their public discourse as a means of loyalty signaling in an authoritarian regime. They find that the state governors in Russia signaled their loyalty to Putin by making more references to him and his political agenda in the addresses to their regional legislatures (Baturo and Khokhlov, 2025).

2.1. Loyalty-signaling paradigm in the political context of Hong Kong

Nevertheless, the political context in Hong Kong does not resemble the ordinary pattern in an authoritarian regime. Whilst the CE and his or her SAR government is accountable to the CPG according to the Basic Law, the relationship between the SAR government and CPG is not strictly one of the lower and higher-level governments in an authoritarian regime because of the high autonomy enjoyed by Hong Kong under the framework of “One Country, Two Systems” (‘OCTS’). For a long time after the 1997 Handover, in order to maintain a high level of stability and continuity in Hong Kong, not causing panics among the businessmen

and ordinary residents in Hong Kong, the CPG took the initiative to refrain from interfering with Hong Kong's political affairs (So, 2011), and gave Hong Kong a very large room of autonomy even in the area of the CPG's prerogatives, such as diplomacy. As regards the election of CE, since it concerns the "sovereignty, security and developmental interest" of the PRC (Chen, 2017, p. 18), the CPG has to exert a certain level of influence to make sure that the CE-elect was not confrontational to the CPG. But before 2020, such an influence, albeit considerable, was indirect and limited. Beijing had to rely largely on its political coalition with the local business elites to secure a favorable result in the CE election, and a non-China-preferred candidate probably would have won the 2017 CE election had this political coalition been shaken (Kwong, 2025). It is thus far-fetched to say that at that time, a CE was hand-picked by the CPG. Hence, it seems that in the pre-2020 Hong Kong, so long as a CE or CE-elect fully respects the CPG's authority and has demonstrated the level of allegiance to the CPG as mandated by the Basic Law, they had little incentive to display additional loyalty through frequent rhetorical references to the central authorities and their political priorities, as such a "sycophantic act" was likely to backfire, undermining the CE's support among ordinary Hong Kong residents and the pan-democrats, who constitute a highly influential political force at that time (Ng, 2020).

But things have changed drastically since 2019. The unprecedentedly large-scale anti-extradition law amendment riots had eroded the CPG's faith in Hong Kong to protect the country's "sovereignty, security and developmental interest" from the pro-independence and pro-separatism groups on its own (Hung, 2020). The landslide victory by pan-democrats and anti-China groups in the 2019 District Council Election has further deepened the CPG's worry about the local regime being 'taken over' by separatist forces in the future elections (Hung, 2020). Consequently, in 2020, Beijing enacted the NSL and promulgated a decision to disqualify some dissenting legislators, followed by a drastic amendment of the methods for electing the Chief Executive (Annex I to the Basic Law) and the Legislative Council (Annex II to the Basic Law) in 2021, which has not only effectively excluded the opposition from accessing Hong Kong's political system but also reconstituted the pro-Beijing coalition—elevating a new tier of loyalist elites while diminishing the proportion of direct election and marginalizing the traditional capitalist class (Kwong, 2025). In this way, Beijing has successfully dominated Hong Kong's executive and legislative systems, becoming the single most important support base for any candidate who wants to run for the CE election and any CE who wants to have effective governance after assuming the office.

With all these changes, now the political context in Hong Kong seems to match the superior-subordinate pattern for the loyalty-signaling theoretical framework as mentioned above. In light of the theoretical framework, the CEs shall have a high incentive to signal their loyalty to Beijing after 2020, as the social unrest in 2019 has weakened Beijing's trust in Hong Kong's politicians and now a CE's political career and survival depend almost solely upon Beijing's endorsement. This theoretical analysis fits well with the phenomenon observed by many people (Li, 2022; Chan, 2023). At the second-half stage of Carrie Lam's administration, her conversation and public pronouncements seemed to mirror Beijing-favored language more frequently, ranging from the reference to political and cultural norms put forward by Beijing, the adoption of Beijing's assertive tone in public addresses to the systematic rebranding of Hong Kong as a subordinate city-region fully integrated into the national development blueprint (Peck et al., 2024). Such a loyalty-signaling discursive practice has been much more normalized since John Lee assumed office in mid-2022. Deng and Lin (2025) point out that citations of

Beijing's official documents and sycophantic declarations of gratitude to Beijing have become standard practices of the incumbent HKSAR government.

2.2. Research Focus

Though existing scholarly literature has touched upon the Hong Kong SAR government's loyalty-signaling discursive practices (Deng and Lin, 2025; Peck et al., 2024; Kwan, 2023), their analyses are far from comprehensive and systemic. Some of them only sample the annual Policy Addresses for analysis, yet it overlooks significant loyalty-signaling rhetoric in other public settings, such as the CE's beginning of his Tai Po Fire media briefings by thanking President Xi for his care and support (Government of the Hong Kong Special Administrative Region, 2025), and the CE's praise of President Xi at the launch of the President's official publications (Government of the Hong Kong Special Administrative Region, 2026). Other existing literature focus predominantly on a concrete policy aspect (Kwan, 2023), which has certainly lost sight of other important dimensions. Also, while measuring the density of keywords with loyalty-signaling features in their quantitative analysis, the existing research fails to provide a comprehensive and systematic list of those keywords, merely arbitrarily selecting them without a substantive explanation on how the inclusion of these keywords in a public speech can signal loyalty to Beijing (Deng and Lin, 2025; Peck et al., 2024).

With the above-mentioned research gaps in mind, our research seeks to systematize the analysis of the CEs' loyalty-signaling discourse mainly by i) compiling and rationalizing a comprehensive and categorized list of keywords used by the CEs to display loyalty to Beijing; ii) sampling a larger and more representative scope of public speeches by the CEs, with the press releases delivered by their offices and the central leaders' Hong Kong-related speeches as supplementary reference data. The methodology for executing these research tasks will be elaborated in detail in the next section.

3. METHODOLOGY

3.1. Data Sampling

The primary dataset consists of 6,994 official press releases and 2,697 speeches and articles in Traditional Chinese, all of which were retrieved from the official websites of the successive CEs. Since these websites serve as a main channel for a CE to communicate with the public and include every significant public address, the data is highly representative of a CE's official voice throughout his or her entire career in office. This corpus spans five administrative terms from July 1997 to March 2026, covering the leadership of Tung Chee-hwa, Donald Tsang, Leung Chun-ying, Carrie Lam, and John Lee.¹

By measuring the density of the Loyalty Signaling Keywords in the corpus, we may find out how the loyalty-signaling discursive practices vary among the five CEs. To facilitate a comparative analysis, addresses by central leaders regarding Hong Kong were collected as a proxy for Mainland political discourse. By evaluating the linguistic similarity between these two corpora, this paper can the degree of rhetorical alignment and discursive convergence between Hong Kong and the CPG.

3.2. Conceptualizing "Loyalty Signaling"

According to the theoretical framework laid out at the beginning, public discourse is an important channel for lower-level officials

¹Press release for current administration (Traditional Chinese) can be found at: <https://www.ceo.gov.hk/tc/press.html>; CE speeches: <https://www.ceo.gov.hk/tc/speeches.html>; Archive of past CE's speeches and press releases: <https://www.ceo.gov.hk/tc/links.html>

to display loyalty to their superiors in an authoritarian regime (Baturu and Khokhlov, 2025), but what to include in the public speeches as loyalty signals might vary in different regime contexts.

Within the sphere of Chinese politics, the concept of ‘loyalty’ is multifaceted and nuanced. While traditional political culture often emphasizes personal allegiance to immediate superiors, the official discourse explicitly prioritizes ‘loyalty to the Party’—a term that remains strategically ambiguous. Does that translate into Hong Kong political discourse as well? How do CEs signal loyalty to the CPG?

This section aims to justify our current categorization of keywords into six types. It is not only a thematic grouping, but a structural mapping based on loyalty-signaling theory. As established by Huang (2015), official propaganda in an authoritarian context often functions not to persuade, but to signal an agent’s submission. The six categories below capture how the Hong Kong government utilizes specific linguistic markers to signal alignment with the central authorities, in order to bridge the “political trust deficit” with Beijing.

To ensure the analysis is comprehensive, these keywords are divided into six categories, conceptually grouped into three layers: Authority (Who), Strategy (What), and Cognition (How).

3.2.1. Authority signals

This layer identifies the source of political mandate and confirms the hierarchy of obedience. This is captured through two primary dimensions: personal loyalty and institutional loyalty.

i. Personal loyalty

The inclusion of keywords related to the top leadership (such as their names and positions) is designed to measure the transition from “rule-based” to “leader-based” legitimacy. In an appointive political hierarchy, such rhetoric serves as “political credentialing.” As suggested by Baturu et al. (2025), reference to the name and political agenda of the top leader is an effective way to signal loyalty in an authoritarian regime. By grounding local initiatives in the “Spirit” or instructions of the “Core” leader, the CE used name projections to signal that the administration’s legitimacy is derived from supreme authority rather than local bureaucratic tradition.

ii. Institutional loyalty

This category tracks keywords related to central party institutions and state organs (e.g., the Central Office of Hong Kong and Macao Affairs, 20th Party Congress). By consistently referencing central institutions, the CE symbolically validates the CPG’s oversight and “plenary power to govern Hong Kong”, proactively acknowledging the principal’s presence and authority.

3.2.2. Strategic alignment

This layer measures the redirection of policy objectives and the reconstruction of Hong Kong’s identity.

iii. Policy Loyalty

As pointed out by Peck et al. (2024), in the wake of the 2019 riot and the NSL in 2020, the CE has changed her discursive emphasis from the unique strength of Hong Kong as an open metropolis and the classical liberal model it relied on to thrive in the past decades, to Hong Kong’s instrumental role in national development and the imperative need of Hong Kong to integrate into the nation’s policy blueprint. Even in the area of climate policy, the SAR government has frequently adopted Mainland government’s framing style in its policy documents since 2019 (Kwan, 2023). Such changes in terms of the policy alignment are undoubtedly

an important discursive practice to express Hong Kong’s subordination to Beijing. In order to capture this discursive paradigm, we set up this category which focuses on keywords that align local governance with national planning (e.g., Great Bay Area, 14th Five-Year Plan). By transforming local administrative tasks into national strategic contributions, Hong Kong government uses performance-based signals to prove they are executing Central Authority’s will. By framing “Hong Kong’s strengths” as instruments for “the country’s needs,” the CE demonstrates functional utility, proving that the SAR is a productive asset in the national development strategy, thereby signaling loyalty to Beijing.

iv. Nationalistic Loyalty

As noted by Kwan (2023), Hong Kong SAR government has repeatedly highlighted its relationship and proximity with Mainland China since 2019, which is also an important category of discourse for signaling loyalty. Keywords like “Integration” [融合] and “Backing the Motherland” [背靠祖国] go beyond simple patriotism, and signify a relational re-anchoring, where Hong Kong is re-defined as a functional dependency of the national whole. By establishing a “mother-child” or “supply-demand” narrative, the Hong Kong government actively signals that Hong Kong’s identity is now inextricably tied to its proximity and contribution to the motherland.

3.2.3. Cognitive convergence

This layer captures the deepest level of alignment, the adoption of the principal’s worldview and linguistic habits.

v. Ideological loyalty

This category includes high-level political concepts (e.g., “Chinese-style Modernization” [中國式現代化], “Struggle Spirit” [鬥爭精神]). The rationale is to determine the extent to which the Hong Kong bureaucracy is adopting the party state’s epistemological framework. This signals that the Hong Kong government is not merely following orders but is actively learning to “think” in synchronization with Beijing.

vi. Rhetorical loyalty

This dimension captures the core of “officialese”, involving mainland-style vocabularies such as “Unswervingly” [堅定不移], “To the letter” [不折不扣], “High-level design” [高屋建瓴]. This represents a form of “Mimetic Isomorphism” (DiMaggio and Powell, 1983), where an organization, facing political uncertainty, adopts the precise communicative rituals of the dominant power to gain legitimacy. By mimicking the rhetorical tempo of the CPG, the CE demonstrates that his or her administration is “fully disciplined,” using symbolic compliance to secure political safety and minimizing the risk of interpretive variance.

3.3. Keyword Detection

The data cleaning phase involved adjusting character encoding for the accurate rendering of Traditional Chinese characters, handling missing values, and stripping HTML tags, punctuation, and extraneous whitespace. A length threshold was enforced, excluding any documents with fewer than 2,000 characters. Additionally, temporal metadata was standardized by extracting years of publication via regular expressions and formatting dates to DD-MM-YYYY, restricting the analytical window strictly to the 1997–2026 period.

Data wrangling procedures include transforming the unstructured corpus into structured DataFrames for quantitative analysis, and conducting text normalization and stop word removal to prepare the text for subsequent processing.

The following computational steps are performed to transform raw text into measurable density metrics:

- Temporal categorization: a regular expression function, `extract_year`, was used to identify the year of publication from the file names.
- Keyword extraction: for every document, the script iterated through the dictionary and counted the total occurrences (Hits) of every defined keyword.
- Normalization and density metrics: to account for varying document lengths, the *Loyalty Signaling Keyword Density* was calculated using the following formula:

$$\text{Density} = \left(\frac{\sum \text{Keyword Hits}}{\text{Total Character Count}} \right) 10,000 \quad (1)$$

This normalization yields a standardized metric representing keyword frequency per 10,000 characters, allowing for a statistically valid longitudinal comparison across the dataset.

3.4. Cosine Similarity Analysis

The CEs' speeches and Hong Kong-related discourse from the central leaders are parsed using regular expressions to extract publication dates and categorized into annual corpora. The raw text undergoes rigorous cleaning, including the removal of non-Chinese characters and HTML tags. The `jieba` library is used for precise Chinese semantic segmentation.

To isolate "loyalty signaling" from routine administrative language, loyalty keyword boosting and customized stop-word filter are implemented. The TF-IDF algorithm is implemented to transform unstructured text into high-dimensional numerical vectors. The model utilizes a `max_df` of 0.85 and a `min_df` of 2 to remove extremely common or statistically insignificant tokens.

The degree of discursive convergence is quantified using Cosine Similarity Formula.

$$\text{Cosine Similarity} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \quad (2)$$

where $\mathbf{A}$ and $\mathbf{B}$ represent the TF-IDF vectors of CE speeches and central leader addresses, respectively. This metric ranges from 0 (completely dissimilar) to 1 (identical), providing a quantitative measure of rhetorical alignment over time.

4. RESULTS AND FINDINGS

4.1. Loyalty Signaling in CE Speeches

The density of loyalty-signaling keywords in CEs' speeches varied significantly across administrations (ANOVA result: $F(4,2692) = 558.02, p < .001$). Tung established a moderate baseline for loyalty-signaling rhetoric, which remained relatively stable throughout his tenure, marked by a slight uptick preceding his resignation. Following Tung, loyalty signaling plummeted. For both Tsang and Leung, the median keyword density dropped to zero, indicating that in at least half of their analyzed speeches, loyalty-signaling keywords did not appear a single time. Although Figure 1 shows a visual upward trend in signaling at the end of Leung's term, statistical analysis demonstrates no significant difference in the overall signaling density between Tsang and Leung (See Table 1).

A dramatic escalation in loyalty signaling occurred during Carrie Lam's administration. The frequency climbed steadily from 2017, reaching its peak in 2020. Her mean loyalty-signaling density was more than double of Tung's and nearly five times higher than Leung's. This rhetorical shift reached its absolute triumph under John Lee's administration. Not only is his keyword density the highest, his lowest Coefficient of Variation (CV) further suggests that his high volume of loyalty signaling is not confined to specific events; it is a standardized, highly consistent, and baked-in component of his everyday executive discourse.

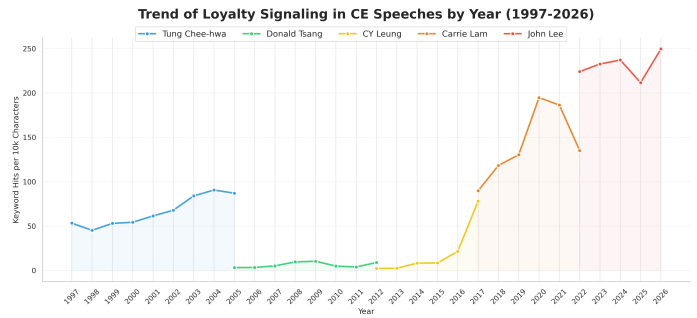


Figure 1. Density of loyalty-signaling keywords across administrations.

Table 1. Statistical Summary and Pairwise Comparison of Loyalty Keyword Density in CE Speeches

CE	n	Descriptive Statistics			Post-hoc Comparison (p-adj)			
		Mean	Median	CV	Tung	Tsang	Leung	Lam
Tung	497	61.15	50.98	0.84	—	—	—	—
Tsang	396	16.20	0.00	2.51	***	—	—	—
Leung	451	29.41	0.00	1.74	***	.20	—	—
Lam	646	139.93	112.40	0.76	***	***	***	—
Lee	707	228.80	211.60	0.56	***	***	***	***

Note. Overall ANOVA: $F(4, 2692) = 558.02, p < .001$.

*** $p < .001$; ** $p < .01$; * $p < .05$.

Chronologically, the year of 2015 serves as a critical inflection point, representing the immediate rhetorical response to the 2014 Umbrella Movement. This shift was preceded by the State Council's June 2014 White Paper, which systemically introduced the doctrine of "Comprehensive Jurisdiction" [全面管治] for the first time.² This trajectory of intensified signaling reached its peak in 2020, following the Anti-Extradition Law Amendment Bill Movement. While the promulgation of the National Security Law ('NSL') in 2020 was followed by a temporary dip in signaling density, the intensity subsequently surged to an all-time high during John Lee's administration. These patterns strongly support the conclusion that loyalty signaling in the HKSAR is most acute during periods of peak political uncertainty and serves as a reparative mechanism to restore central-local confidence.

While the overall trend for "loyalty signaling" shows a statistically significant increase during the tenures of Carrie Lam and John Lee, this pattern is not uniform across all categories (as suggested by Figure 2). First of all, there is no rhetorical loyalty signaling throughout all the periods. Secondly, ideological loyalty has constantly remained the most dominant category in terms of absolute volume, reaching a density of 120.4 per 10,000 characters in Lee's administration. But it shall be noted that Tung Chee-hwa also exhibited a high propensity for ideological signaling; as the first Chief Executive, he had to frequently articulate the relationship between the HKSAR and the mainland socialist system. Thirdly, none of the five CEs demonstrated a significant enthusiasm for "personal loyalty" signaling, indicating that political alignment in the HKSAR is projected as an institutional or systemic commitment rather than an individual allegiance.

²See https://www.hmo.gov.cn/xwzx/zwyw/202312/t20231208_38591.html This document signaled Beijing's dissatisfaction with the local administration's perceived inability to manage the precursors to "Occupy Central," emphasizing that Hong Kong's autonomy is a delegated power rather than an inherent right.

Loyalty Keyword Density Across Chief Executive Tenures (1997-2026)

Values represent average occurrences of loyalty-signaling terms per 10,000 characters in CE speeches.

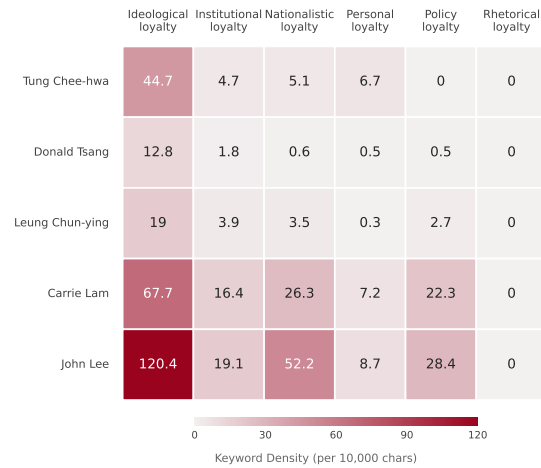
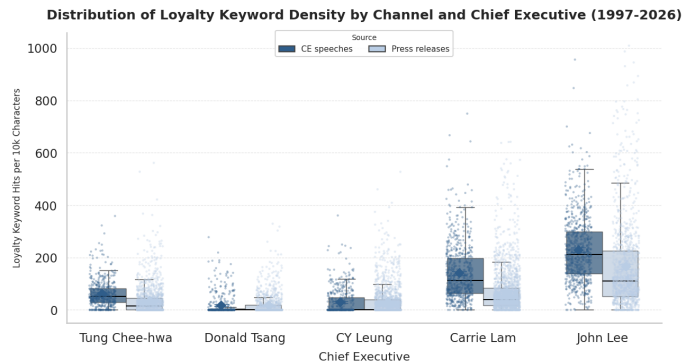


Figure 2. Density of loyalty-signaling keywords by category across administrations.

4.2. Loyalty Signaling in Press Releases

For Tung Chee-hwa, Carrie Lam, and John Lee, there is a statistically significant difference ($p < .001$) between their spoken speeches and written press releases. In all three cases, they deploy vastly more loyalty signaling through spoken words than in press releases. Whereas for Tsang and Leung, such gap is absent. The Welch t -test results indicate no statistically significant difference between what they said and what their administrations published. When overall loyalty signaling was low, it was uniformly low across both channels.



Note. Diamonds show mean with 95% CI (bootstrap, 1,000 resamples).
Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. $p > 0.05$.

Figure 3. Comparison of loyalty-signaling keyword density between CE speeches and press releases.

Table 2. Comparison of Loyalty Keyword Density (Welch's t -test)

CE	Speech n	Press n	Mean (S)	Mean (P)	t -stat	p -value
Tung	499	1338	60.90	34.46	9.60	< .001***
Tsang	423	1105	17.43	17.39	0.02	.99
Leung	451	1603	29.41	31.47	-0.74	.46
Lam	646	1878	139.93	67.74	15.59	< .001***
Lee	707	1070	228.80	165.14	9.02	< .001***

Note. Mean (S) = Speeches; Mean (P) = Press Releases.
Welch's t -test was used due to unequal variances and sample sizes.
*** $p < .001$; ** $p < .01$; * $p < .05$.

There is a notable shift in the Lee administration's loyalty signaling channel turning into 2026 (See Figure 6 in Appendix B). For the first time since 1997, the data reveal an inversion in the

signaling medium, with significantly more loyalty being signaled through official press releases than through the Chief Executive's personal speeches. While it remains unclear whether this marks an outlier or a strategic realignment in public discourse, two hypotheses offer potential explanations for this phenomenon. The first hypothesis is that, having spent 2022 to 2025 establishing an undeniably high baseline of political alignment, the administration may now consider that trust secured. The burden of loyalty signaling is being pushed down into the civil service machinery through the standardized, boilerplate text of daily press releases. Another hypothesis is that the administration, cautious of potential backlash from civil society and the commercial sector—where repeated loyalty signaling might shake confidence in Hong Kong's autonomy—the administration may be intentionally shifting these signals into less visible bureaucratic channels. This allows the Chief Executive to reserve his or her highly visible public speeches for pragmatic, economically oriented messaging.

4.3. Textual Similarity

The consistently low overall textual similarity between Chief Executive speeches and state leaders' Hong Kong-related addresses highlights a crucial finding: loyalty signaling has not driven a wholesale assimilation into Mainland political rhetoric (See Figure 7 in Appendix B). This demonstrates that the recent rise in loyalty signaling has not fundamentally changed the underlying structure of Hong Kong's political discourse. Despite a higher frequency of specific political keywords, there is no empirical evidence to suggest that the CE has adopted the generalized "officialese" often characteristic of Mainland bureaucratic texts.

Notably, contrary to the hypothesis that John Lee might heavily utilize Mainland-style bureaucratic jargon to approximate Beijing's political discourse, it appears from the data that he employs a highly targeted, "modular" signaling strategy. Rather than structurally adopting Mainland-style rhetorical style, Lee strategically inserts specific loyalty-signaling keywords (such as "national security" and "patriots administering Hong Kong") with high frequency and precision to establish an unambiguous political baseline. However, his overarching narrative style, daily administrative vocabulary, and linguistic structure have not converged with the Mainland's official speech system. This represents a highly pragmatic rhetorical approach: it guarantees strict political compliance while actively preserving the localized, pragmatic, and technocratic linguistic framework inherent to Hong Kong's executive branch.

5. CONCLUSION AND DISCUSSION

This paper has elucidated the dynamic evolution of loyalty-signaling discursive practices by Hong Kong's Chief Executives from 1997 to early 2026, revealing that political alignment is a strategic tool rather than a constant or permanent posture. The findings demonstrate that loyalty signaling serves as a reparative mechanism, intensifying significantly following periods of socio-political friction, such as the 2014 Umbrella Movement and the 2019 Anti-Extradition Law Amendment Movement, to regain political confidence from the Central Government. While the frequency of such signals remained moderate or low during the terms of Tung Chee-hwa, Donald Tsang, and Leung Chun-ying, it experienced a dramatic surge under Carrie Lam and reached a peak of standardized consistency during John Lee's administration. However, the data refutes the hypothesis of a wholesale assimilation, namely fully adopting the mainland rhetorical styles; instead, it reveals a "modular" strategy where specific political baselines are established through high-precision keywords while the underlying technocratic and pragmatic features of HKSAR government's official discourse remain preserved. The observed

shift in 2026, where signaling density in official press releases surpassed that of personal speeches, further suggests a transition towards the institutionalization of loyalty within the bureaucratic apparatus. In this more invisible manner, the CE and his administration may signal loyalty to Beijing without undermining the faith of businessmen and residents in the “high autonomy” of Hong Kong under the OCTS framework. With the full picture of results in mind, it seems that the political elites in Hong Kong, represented by the CE, generally do not signal loyalty to Beijing for its own sake; rather, they skilfully express loyalty with the aim of reassuring Beijing that the “Two Systems” element would not undermine the “One Country” component under the OCTS framework, so that they can gradually regain more political room for the SAR. When Beijing rests fully assured of the loyalty of Hong Kong’s political elites, the democratization process as outlined and envisaged in the Basic Law might go back on the political agenda.

However, several limitations of this research must be acknowledged. First, the analysis relies exclusively on public speeches and press releases, which may represent a “front-stage” performance that does not fully capture the complexities of private, “back-stage” interactions between SAR officials and Beijing. Second, since we used keyword detection and text similarity as the major methodologies, despite the fact that the quantitative metrics provide a robust macro-level overview, they may overlook the subtle linguistic nuances, tone, or coded metaphors that characterize political communication beyond specific keyword densities. Finally, the focus on Chinese-language documents leaves a gap in understanding how the CEs manage multi-audience signaling, particularly when addressing international stakeholders in English.

Despite these constraints, this research contributes to the existing literature by enriching the academic understanding of the loyalty-signaling discursive paradigm in semi-autonomous contexts and systematizing the analysis of loyalty-displaying discourses in Hong Kong. By developing a six-dimensional keyword framework to concretize “loyalty,” this study moves beyond narrating qualitative anecdotes to providing a measurable and empirical basis for analyzing political alignment. Methodologically, the application of text mining and semantic similarity analysis offers a replicable model for studying political discourse in other Chinese provinces or autonomous regions. Looking forward, this research provides a foundation for future inquiries into the audience reception of modular signaling, particularly among the business sector and international community, regarding whether there are significant audience costs of such high-density loyalty rhetoric. Future studies could also expand this framework to investigate whether the current linguistic balance is a sustainable equilibrium or a transitional phase that will eventually converge with mainland political rhetoric.

Notes on AI Use

The authors utilized AI tools for data cleaning, keyword extraction, and preliminary analysis. However, all final interpretations, conclusions, and the writing of this report were conducted independently by the authors without AI assistance. The AI was used solely as a tool for processing and organizing data, and did not contribute to the intellectual content or narrative of the report.

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■ APPENDIX

A. Loyalty Signaling Keywords

The keywords used to capture loyalty signaling are categorized into the following six dimensions:

1. Personal Loyalty: 習近平, 習近平總書記, 國家主席習近平, 習主席, 國家主席, 總書記, 中共中央總書記, 中共中央總書記、國家主席習近平, 黨和國家領導人, 中央領導, 中央領導人, 國家領導人, 國務院總理, 總理, 國務院副總理, 副總理, 夏寶龍, 夏寶龍主任, 韓正, 韓正副總理, 丁薛祥, 丁薛祥副總理, 李強, 李克強, 胡錦濤, 胡主席, 江澤民, 江主席, 溫家寶, 朱鎔基, 李鵬, 鄧小平, 港澳辦主任, 中央有關部委, 中央相關部委。

2. Institutional Loyalty: 中央人民政府, 中央政府, 黨中央, 國務院, 國務院港澳事務辦公室, 港澳辦, 中央港澳工作辦公室, 中聯辦, 中央人民政府駐香港特別行政區聯絡辦公室, 中央人民政府駐香港特別行政區維護國家安全公署, 外交部, 商務部, 國家發展和改革委員會, 國家發改委, 全國人民代表大會, 全國人大, 全國人大常委會, 人大常委會, 全國政協, 港區全國人大代表, 港區全國政協委員, 全國兩會, 兩會, 全國兩會精神, 人民大會堂, 中國共產黨, 中央委員會, 中央政治局, 中共中央政治局, 中央軍委, 二十大, 中國共產黨第二十次全國代表大會, 二十大精神, 十九大, 中國共產黨第十九次全國代表大會, 十九大精神, 十八大, 中國共產黨第十八次全國代表大會, 十八大精神, 十七大, 十六大, 十五大, 二十屆三中全會, 中共二十屆三中全會, 二十屆三中全會精神, 三中全會, 三中全會精神, 四中全會, 五中全會, 六中全會, 維護國家安全委員會, 香港特別行政區維護國家安全委員會, 國安委, 駐港國家安全公署。

3. Policy Loyalty: 一帶一路, 一帶一路倡議, 共建一帶一路, 粵港澳大灣區, 粵港澳大灣區建設, 大灣區建設, 粵港澳大灣區發展規劃綱要, 十四五, 十四五規劃, 十四五規劃綱要, 國家十四五規劃, 十三五, 十二五, 十一五, 十五五, 五年規劃, 五年規劃綱要, 高質量發展, 新質生產力, 新發展格局, 新發展理念, 創新驅動發展, 供給側結構性改革, 共同富裕, 國內國際雙循環, 雙循環, 高水平對外開放, 高水平安全, 北部都會區, 北部都會區行動綱領, 河套深港科技創新合作區, 河套深港科技創新合作區深圳園區發展規劃, 河套合作區, 前海深港現代服務業合作區, 橫琴粵澳深度合作區, 國際創新科技中心, 愛國主義教育, 國家安全教育。

4. Nationalistic Loyalty: 融入國家發展, 融入國家發展大局, 積極融入國家發展大局, 更好融入國家發展大局, 更好融入和服務國家發展大局, 服務國家發展大局, 支持香港更好融入國家發展大局, 積極主動融入國家發展大局, 主動積極融入國家發展大局, 全面融入國家發展大局, 更積極融入國家發展大局, 推動香港更好融入國家發展大局, 背靠祖國、聯通世界, 背靠祖國, 聯通世界, 對接國家戰略, 對接國家發展戰略, 服務國家戰略, 服務國家所需, 國家所需, 貢獻國家所需, 貢獻國家發展, 香港所長, 發揮香港所長, 以香港所長貢獻國家所需, 說好香港故事, 講好香港故事, 說好中國故事, 說好國家故事, 前海, 南沙, 河套, 橫琴, 深化粵港澳合作, 積極對接國家戰略, 主動對接國家戰略, 內聯外通, 內聯外通優勢, 全面準確貫徹一國兩制, 全面準確貫徹一國兩制方針, 一國兩制實踐, 一國兩制事業, 長期堅持一國兩制, 港人治港, 愛國者治港, 愛國者治港原則, 由治及興, 國家安全, 維護國家安全, 總體國家安全觀, 憲法, 基本法, 憲法和基本法, 憲制秩序, 憲制責任, 全面管治權, 完善選舉制度, 愛國主義, 愛國, 愛國愛港, 愛國者, 國家觀念, 國家意識, 國民身分認同, 家國觀念, 正本清源, 撥亂反正, 反中亂港, 外部勢力, 依法治港, 維護國家主權, 維護國家主權、安全、發展利益, 國家主權, 國家主權、安全、發展利益, 安全和發展利益, 國家利益, 國家安全教育日, 全民國家安全教育日, 香港國安法, 國安法, 國安條例, 基本法第二十三條, 二十三條立法。

5. Ideological Loyalty: 中國特色社會主義, 中國特色社會主義制度, 新時代中國特色社會主義, 習近平新時代中國特色社會

主義思想, 新時代中國特色社會主義思想, 習近平文化思想, 鄧小平理論, 三個代表重要思想, 科學發展觀, 中國式現代化, 強國建設, 社會主義現代化強國, 現代化強國, 中華民族偉大復興, 民族復興, 中華民族共同體, 中華民族共同體意識, 鑄牢中華民族共同體意識, 中國夢, 家國情懷, 愛國精神, 愛國主義精神, 中華文化, 中華優秀傳統文化, 文化自信, 制度優勢, 制度自信, 道路自信, 理論自信, 馬克思主義, 黨的領導, 黨的全面領導, 兩個確立, 兩個維護, 四個意識, 四個自信, 全過程人民民主, 偉大建黨精神, 自我革命, 底線思維, 憂患意識, 鬥爭精神, 人民至上, 生命至上, 政治站位, 政治判斷力, 政治領悟力, 政治執行力, 合作共贏, 多邊主義, 人類命運共同體, 社會主義核心價值觀。

6. Rhetorical Loyalty: 重要講話, 重要講話精神, 習近平主席重要講話, 國家主席重要講話精神, 重要回信, 重要回信精神, 回信精神, 賀信, 賀信精神, 重要指示精神, 指示精神, 批示精神, 重要指示批示, 大力支持, 全力支持, 明確支持, 充分支持, 在中央的大力支持下, 在中央政府的大力支持下, 中央政府的大力支持, 中央支持, 中央對香港的支持, 中央對香港的關愛, 中央充分肯定, 中央高度重視, 中央高度關注, 高度重視, 高度肯定, 充分肯定, 充分認可, 充分信任, 親切關懷, 親切關心, 親切接見, 親切慰問, 殷殷囑託, 殷切關懷, 殷切期望, 殷切期望, 巨大鼓舞, 倍感振奮, 深刻指出, 指明方向, 高屋建瓴, 精闢論述, 全面準確貫徹, 貫徹落實, 堅定不移, 貫徹, 落實, 推進, 深化, 完善, 健全, 統籌, 部署, 強化, 優化, 鞏固, 提升, 構建, 對接, 融入, 協同, 扎實, 紮實, 切實, 積極, 主動, 全面, 深入, 高質量, 高水平, 加快, 全力, 持續, 進一步, 着力, 配合, 支撐, 推動, 保障, 開展, 實施, 學習, 學習宣傳貫徹, 深入學習貫徹, 研習, 研討, 宣講, 傳達, 領會精神, 行政主導, 奮發有為, 主動作為, 擔當作為, 先立後破, 以進促穩, 穩中求進, 守正創新, 善作善成, 敢於擔當, 夯實基礎, 紮實推進, 縱深推進, 全面貫徹落實, 堅決貫徹落實, 不折不扣, 走深走實, 以結果為目標, 因地制宜, 提速提效, 提質增效。

B. Figures

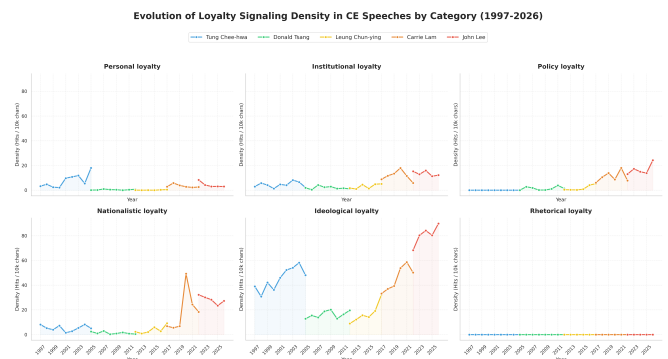


Figure 4. Density of loyalty-signaling keywords in CE speeches by category.

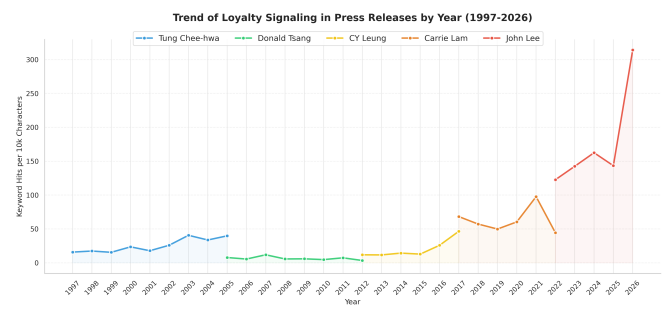


Figure 5. Density of loyalty-signaling keywords in press releases by year.

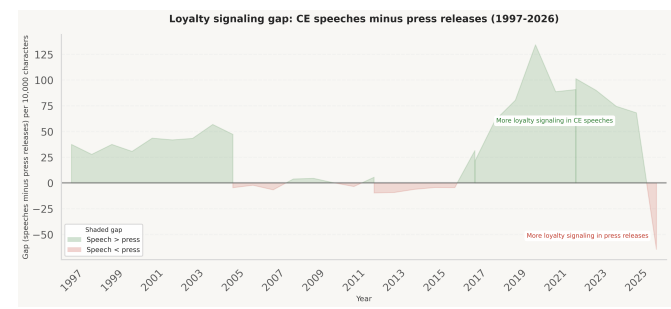


Figure 6. Gap in loyalty signaling between CE speeches and press releases.

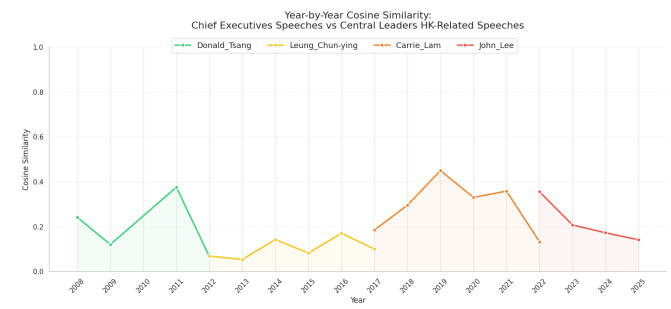


Figure 7. Temporal trend of cosine similarity between CE speeches and central leader addresses.

C. Tables

Table 3. Statistical Summary and Pairwise Comparison of Loyalty Key-word Density in Press Releases

CE	n	Descriptive Statistics			Post-hoc Comparison (<i>p</i> -adj)			
		Mean	Median	CV	Tung	Tsang	Leung	Lam
Tung	1338	34.46	15.44	1.63	—	—	—	—
Tsang	1105	17.38	0.00	2.20	***	—	—	—
Leung	1603	31.47	0.00	1.80	.90	***	—	—
Lam	1878	67.74	38.95	1.27	***	***	***	—
Lee	1070	165.14	110.81	1.03	***	***	***	***

Note. Overall ANOVA: $F(4, 6989) = 506.68, p < .001$.
*** $p < .001$; ** $p < .01$; * $p < .05$.

Table 4. Statistical Summary and Pairwise Comparison of Cosine Similarity in CE Speeches

CE	n	Descriptive Statistics			Welch's <i>t</i> -test <i>p</i> -value			
		Mean	Median	Std Dev	Tsang	Leung	Lam	Lee
Tsang	182	0.08	0.03	0.10	—	—	—	—
Leung	448	0.04	0.03	0.04	***	—	—	—
Lam	646	0.09	0.06	0.09	.35	***	—	—
Lee	671	0.06	0.04	0.06	.02*	***	***	—

Note. Significance levels calculated with unequal variances.
*** $p < .001$; ** $p < .01$; * $p < .05$.

GITHUB REPOSITORY

Visit the repository to access the source code, track ongoing development, report issues, and stay up to date with the latest changes.

 https://github.com/dengsherry2003-ui/Final_Project